EXPERIMENT 10

APPLYING PERFORMANCE OPTIMIZATION TECHNIQUES IN ZOHO CREATOR APPLICATION

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PREPARED BY

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1 Aim

> **AIM** To demonstrate performance optimization techniques in a Zoho Creator Event Registration application by retrieving only the required record and improving the efficiency of workflow processing.

The experiment focuses on reducing unnecessary record retrieval and unnecessary processing inside a form workflow.

2 Introduction

A low-code platform such as Zoho Creator takes care of much of the underlying work of running an application, yet the way a workflow or script is written still affects how efficiently it runs. In this setting, performance optimization means designing forms, workflows and Deluge scripts so that they use only the data and the processing that a task actually requires.

When a script retrieves more records than it needs, or repeatedly processes records that have no bearing on the task, it increases the amount of data handled and the amount of work performed. This may go unnoticed in a small application, but as the number of records and users grows it can reduce the efficiency of the application. For this reason, requesting the single record needed for the current operation is preferable to fetching an entire collection and searching through it.

A workflow that retrieves the required record, checks the one field it depends on and then performs a single action is also easier to read and maintain, because there is less logic to follow and no loop to interpret. Such a design supports applications that remain scalable and maintainable.

3 Objectives

After completing this experiment, the student should be able to:

1. Explain what performance optimization means in Zoho Creator.
2. Recognise unnecessary record retrieval and iteration in a Deluge script.
3. Retrieve the current registration record by using its ID.
4. Validate the attendee's email address before a notification is sent.
5. Reduce unnecessary processing through criteria-based record retrieval.
6. Describe how efficient workflow design can improve application logic.

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4 Software and Tools Required

Requirement	Role in this experiment
Zoho Creator	Low-code platform in which the application and its workflow are opened and examined.
Event Registration application	The application used to demonstrate the optimization.
Deluge scripting	Scripting language used to write the logic of the form workflow.
Form workflows	Workflow type in which the Send registration email workflow is defined.
Successful form submission trigger	Event that starts the workflow once a registration is submitted successfully.
Email notification functionality	The sendmail task that sends the confirmation message to the attendee.

5 Application Overview

The experiment is carried out on an Event Registration application built in Zoho Creator. The application contains a form for event registration. After a registration has been submitted successfully, a form workflow processes the current registration record and attempts to send a confirmation email to the attendee.

The logic of the application can be followed through this sequence:

1. An attendee submits the registration form.
2. The form workflow is triggered after the successful submission.
3. The workflow identifies the current record by its ID.
4. The relevant registration record is retrieved with a criteria-based query.
5. The workflow checks whether the email address is blank.
6. If an email address is available, a confirmation email is sent.
7. If the email address is missing, an informational message is displayed.

6 Theory and Performance Optimization Concepts

The central optimization in this experiment is criteria-based record fetching: instead of retrieving every record of a form, a criterion is supplied so that only the required record is returned.

6.1 Unnecessary record retrieval

If every record of a form is retrieved when only one is required, the script handles data that the task never uses. Each additional record adds to the data retrieved and processed, so the cost of the unnecessary work grows along with the number of registrations.

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6.2 Criteria-based record retrieval

By using the ID of the current record as a selection criterion, the script can request the relevant registration directly. The workflow uses the following expression:

```
eventRegistration = Event_Registration[ID == input.ID];
```

Element	Role in the expression
`Event_Registration`	The Event Registration form, whose records are being queried.
`ID`	The ID of each record in the form, used on the left side of the criterion.
`input.ID`	The ID of the record being processed, which belongs to the current form submission.
`eventRegistration`	The variable that receives the single record whose ID matches input.ID.

6.3 Avoiding unnecessary iteration

Fetching the whole collection and looping through it to find the current record is unnecessary when the record can be selected directly with a criterion. The direct approach removes the loop and the repeated comparison of record IDs, and the intent of the statement is clear at a glance.

6.4 Input validation

The confirmation email is sent to the address stored in the retrieved record. The workflow therefore checks whether the email field is blank before attempting to send. When the field is empty there is no address to send to, so the script displays an informational message and does not proceed to the email step.

7 Procedure

Follow the steps below in the Event Registration application. Each step corresponds to a stage of the workflow described in the previous sections.

Step 1 — Open the Zoho Creator application

Sign in to Zoho Creator and open the Event Registration application. Open it in **Edit mode** so that the design and workflow configuration of the application are available.

Step 2 — Open the workflow configuration

In Edit mode, go to the **Workflow** section and open **Form workflows**. The form **Event Registration** is listed under **Forms with workflows**, which shows that at least one workflow is attached to it.

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Step 3 — Locate the Send registration email workflow

Select the Event Registration form and look for the workflow named **Send registration email**. Its configuration is as follows (Figure 1):

Workflow name	Send registration email
Form	Event Registration
Action	Created
Trigger	Successful form submission
Details	1 Action
Status	Enabled

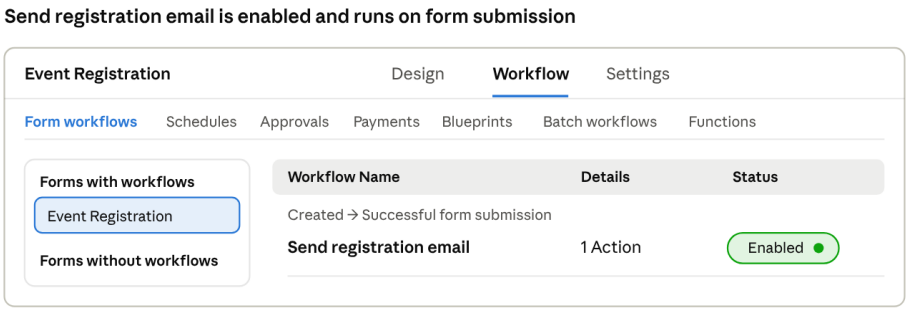


Figure 1: *Form workflow of the Event Registration application (illustrative representation, not a screenshot)*

Step 4 — Identify the current record

When an Event Registration is submitted, the workflow needs to know which record to work with. It uses `input.ID`, which holds the ID of the record associated with the current form submission.

Step 5 — Retrieve the required record

The workflow then fetches that record with a criteria-based statement:

```
eventRegistration = Event_Registration[ID == input.ID];
```

This expression returns the registration whose ID matches the ID of the current submission. The script does not fetch all registrations and search through them manually, so only the relevant record is retrieved.

Step 6 — Validate the email address

Next, the workflow checks the email field of the retrieved record with this condition:

```
eventRegistration.email.isBlank()
```

If the email address is blank, the workflow displays the informational message "Email address is missing for this registration". Otherwise, it proceeds to send the confirmation email.

Step 7 — Send the confirmation email

When an email address is available, the workflow uses `sendmail` to send the notification to the address stored in the retrieved registration record. The sender is `zoho.adminuserid` and the subject of the email is **Your Event Registration is Confirmed**.

The message addressed to the attendee informs them that their registration for the event has been received, says that the organisers look forward to seeing them, and thanks them for registering.

Step 8 — Review the optimized workflow

Review the workflow as a whole: identify the current record, retrieve it by its ID, validate the email field, and then send the email or display the message (Figure 4 in Section 11). Because the record is selected with a criterion, the workflow avoids the full-collection loop shown in the inefficient example of Section 9, in which every record is retrieved and then examined one by one.

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8 Deluge Script

The complete Deluge script used in the Send registration email workflow is given below.

```
eventRegistration = Event_Registration[ID == input.ID];

if(eventRegistration.email.isBlank())
{
    info "Email address is missing for this registration";
}
else
{
    sendmail
    [
        from :zoho.adminuserid
        to :eventRegistration.email
        subject :"Your Event Registration is Confirmed"
        message :"<p>Dear Attendee,</p><p>Your registration for the event has
been received. We look forward to seeing you.</p><p>Thank you for
registering.</p>"
    ]
}
```

8.1 Purpose of the main components

Code component	Purpose
`input.ID`	Identifies the current record associated with the submission.
`Event_Registration[ID == input.ID]`	Retrieves the registration record matching the specified ID.
`eventRegistration.email.isBlank()`	Checks whether the email field is blank.
`info`	Displays an informational message when the email is missing.
`sendmail`	Sends the confirmation email when the email address is available.
`zoho.adminuserid`	Specifies the sender using the configured Zoho administrator user ID.

9 Performance Comparison

The optimized approach used in the workflow is compared below with a general inefficient approach. In the inefficient approach the steps are: fetch all records, iterate through them, find the required record and then process it. In the optimized approach the required record is fetched using its ID and processed directly.

Aspect	Inefficient approach	Optimized approach
Record retrieval	Retrieves the full collection of registration records.	Retrieves only the record whose ID matches `input.ID`.
Need for iteration	A `for each` loop is needed to locate the required record.	No explicit loop is needed; the record is selected by its criterion.
Unnecessary processing	Records that are not needed are retrieved and examined.	Unrelated records are neither retrieved nor iterated through.
Code complexity	Longer, with a loop and a nested condition.	A single criteria-based statement, so the logic stays short and direct.
Suitability when only the current record is required	Less suitable, because the loop handles records beyond the one required.	Well suited, because it targets the current record directly.

Figure 2 illustrates the difference in the number of records retrieved by the two approaches.

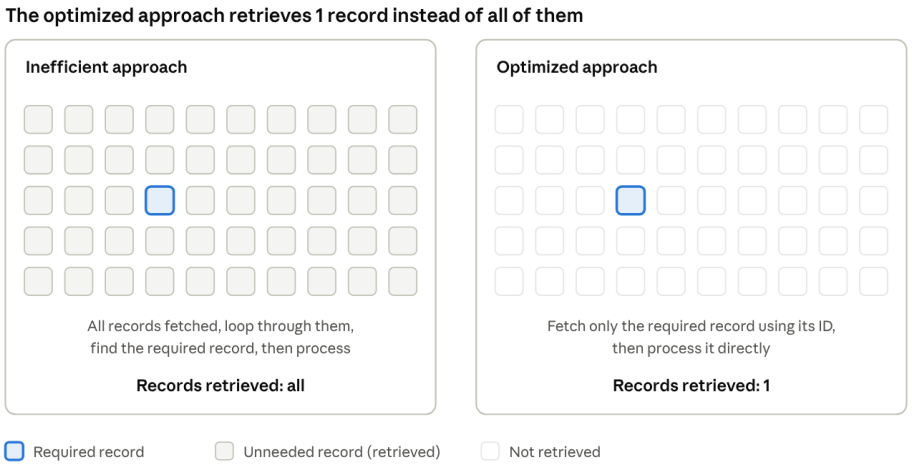


Figure 2: *Records retrieved by the inefficient and optimized approaches (conceptual diagram)*

9.1 Inefficient approach (illustrative example)

The following code is an illustrative example from the comparison. It was not used in the application, and the ellipsis and comment are placeholders rather than a working script.

```
eventRegistrations = Event_Registration[];
for each rec in eventRegistrations
{
    if(rec.ID == input.ID)
    {
        // ...
    }
}
```

This approach retrieves the full collection of records and then iterates through it to locate the required record, so records that are not needed are retrieved and processed.

9.2 Optimized approach (used in the workflow)

```
eventRegistration = Event_Registration[ID == input.ID];
```

This approach selects the record directly through its ID criterion, which avoids the explicit full-collection loop of the inefficient example.

10 Optimization Techniques

The source describes five general practices that contribute to performance in Zoho Creator applications. Not all of them are implemented in this experiment, so the table states the status of each one.

No.	Technique	Purpose	Status
1	Criteria-based fetching	Retrieve only the records required for the current operation, here the registration identified by its ID.	**Used in this practical**
2	Avoiding unnecessary iteration	Do not fetch all records and loop through them when a criterion can identify the required record.	**Followed**
3	Efficient built-in functions and aggregate operations	Use suitable built-in functions and aggregate operations instead of unnecessary repeated processing.	**General practice**
4	Indexed fields	Indexed fields can improve retrieval of frequently queried data. Indexing may require backend or platform configuration, and it was not enabled or configured in this application.	**Not configured**
5	Efficient workflows	Keep workflows simple: retrieve the required record, validate the necessary field and perform the relevant action, without nested or repeated operations.	**General practice**

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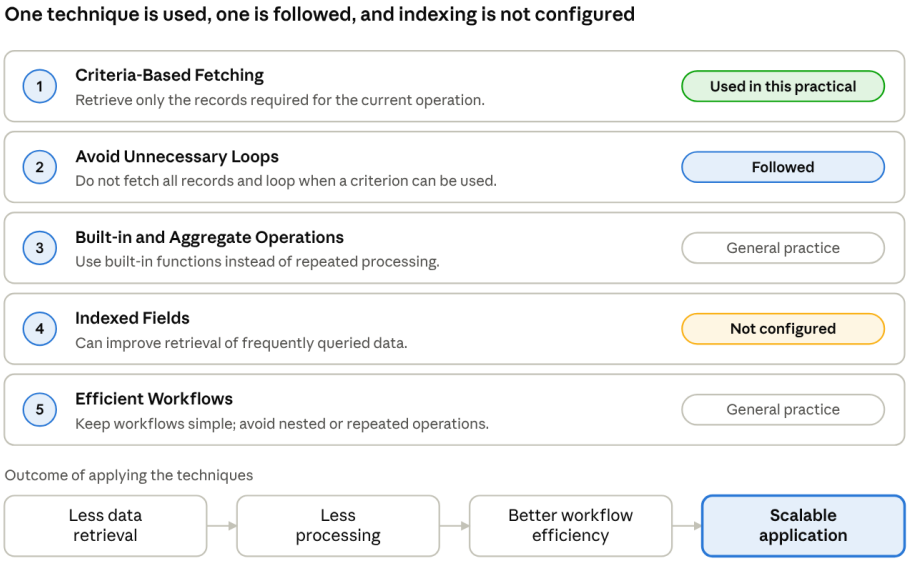


Figure 3: *Performance optimization techniques (conceptual diagram). Outcome: less data retrieval → less processing → better workflow efficiency → scalable application*

11 Optimized Workflow

Figure 4 shows the optimized Event Registration workflow. It follows the logic of the script exactly, with the criteria-based fetch placed before the email check so that only the relevant record is handled.

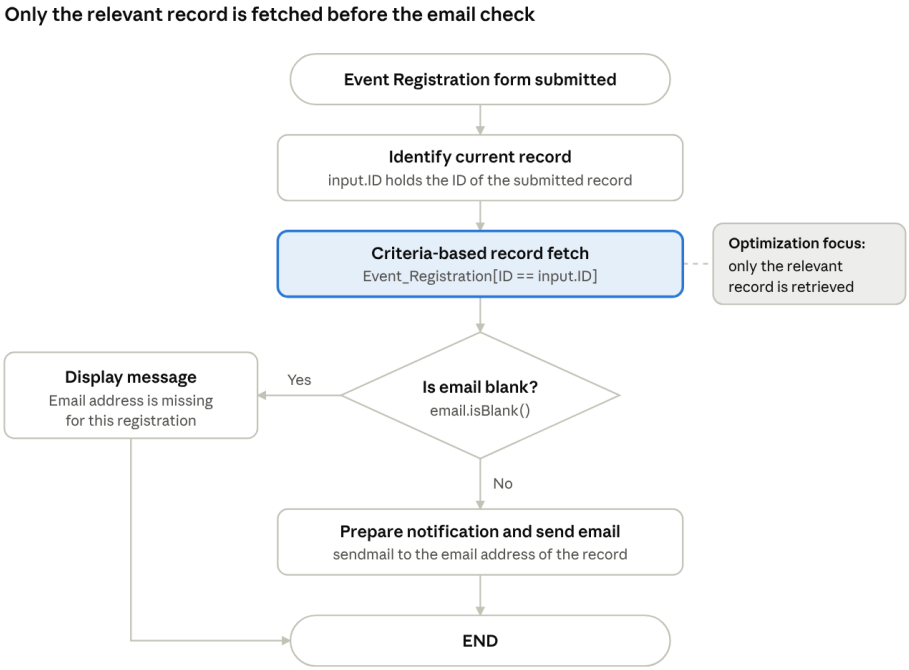


Figure 4: *Optimized Event Registration workflow (conceptual diagram)*

The stages of the flowchart are described below:

1. **Identify the current record:** `input.ID` gives the ID of the submitted record.

2. **Criteria-based record fetch:** `Event_Registration[ID == input.ID]` retrieves only the relevant record.
3. **Validate the email:** `email.isBlank()` checks whether the email field is empty.
4. **Prepare the notification and send the email:** if the email is available, the message is prepared and sent with `sendmail`; otherwise an informational message is displayed.

13 Conclusion

This experiment showed that focused record retrieval and a clearly structured workflow can avoid unnecessary data processing in a Zoho Creator application. By selecting the current registration with `ID == input.ID`, the workflow did not have to retrieve unrelated records or loop through them, and it kept the logic short and direct.

Checking the email field before sending the notification also helps the workflow handle registrations that have no email address, since an informational message is displayed instead of an attempt to send. Together, these techniques give a more direct and maintainable approach to the demonstrated workflow. As no benchmark was carried out, the benefits are expressed in terms of reduced data retrieval and processing, not as measured execution times.

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